Math 5601 - Independent Study Report

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Proposed Topic: *Adaptive Quadrature Methods for Numerical Integration*

1 Integration of Smooth Function: $f(x) = \sin(x)$ on $[0, 1]$

Exact Value (J_F): _____

Parameters: $a = 0$, $b = 1$. Adaptive tolerance: $\epsilon = \delta = 10^{-10}$.

Table 1: Results for Smooth Function $f(x) = \sin(x)$

Method	**N / Type**	$I_{\textbf{approx}}$ **(Approximate Value)**	$\lvert E \rvert$ **(Absolute Error)**
Simple Simpsons Rule			
Simple 3/8 Simpsons Rule			
Composite Integration (Fixed N)			
Composite: Step $N = 5$			
Composite: Step $N = 10$			
Composite: Step $N = 20$			
Composite: Step $N = 50$			
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}: _____		
Adaptive 3/8 Simpsons Rule	N_{final}: _____		

2 Integration of Rapidly Changing Function: $f(x) = \frac{1}{1+100x}$ on $[0, 1]$

Exact Value (J_F): _____
Parameters: $a = 0$, $b = 1$. Adaptive tolerance: $\epsilon = \delta = 10^{-10}$.

Table 2: Results for Rapidly Changing Function $f(x) = 1/(1 + 100x)$

Method	**N / Type**	$I_{\textbf{approx}}$ **(Approximate Value)**	$\lvert E \rvert$ **(Absolute Error)**
Simple Simpsons Rule			
Simple 3/8 Simpsons Rule			
Composite Integration (Fixed N)			
Composite: Step $N = 5$			
Composite: Step $N = 10$			
Composite: Step $N = 20$			
Composite: Step $N = 50$			
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}: _____		
Adaptive 3/8 Simpsons Rule	N_{final}: _____		

3 Integration of Locally Unsmooth Function: $f(x) = \sqrt{x^2 + A}$

Parameters: $a = -1$, $b = 1$.
The table below tracks the error as parameter A shrinks from 10^{-1} to 10^{-10}.

For A = 10^{-1}
Exact Value (J_F): _____

Table 3: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$

Method	**N / Type**	$I_{\mathbf{approx}}$ **(Approximate Value)**	$\lvert E \rvert$ **(Absolute Error)**
Simple Simpsons Rule			
Simple 3/8 Simpsons Rule			
Composite Integration (Fixed N)			
Composite: Step $N = 5$			
Composite: Step $N = 10$			
Composite: Step $N = 20$			
Composite: Step $N = 50$			
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}:		
Adaptive 3/8 Simpsons Rule	N_{final}:		

For A = 10^{-2}
Exact Value (J_F): _____

Table 4: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$

Method	**N / Type**	$I_{\textbf{approx}}$ **(Approximate Value)**	$\lvert E \rvert$ **(Absolute Error)**
Simple Simpsons Rule			
Simple 3/8 Simpsons Rule			
Composite Integration (Fixed N)			
Composite: Step $N = 5$			
Composite: Step $N = 10$			
Composite: Step $N = 20$			
Composite: Step $N = 50$			
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}:		
Adaptive 3/8 Simpsons Rule	N_{final}:		

For A = 10^{-3}
Exact Value (J_F): _____

Table 5: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$

Method	**N / Type**	$I_{\textbf{approx}}$ **(Approximate Value)**	$\lvert E \rvert$ **(Absolute Error)**
Simple Simpsons Rule			
Simple 3/8 Simpsons Rule			
Composite Integration (Fixed N)			
Composite: Step $N = 5$			
Composite: Step $N = 10$			
Composite: Step $N = 20$			
Composite: Step $N = 50$			
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}:		
Adaptive 3/8 Simpsons Rule	N_{final}:		

For A = 10^{-4}

Exact Value (J_F): _____

Table 6: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$

Method	**N / Type**	$I_{\textbf{approx}}$ **(Approximate Value)**	$\lvert E \rvert$ **(Absolute Error)**
Simple Simpsons Rule			
Simple 3/8 Simpsons Rule			
Composite Integration (Fixed N)			
Composite: Step $N = 5$			
Composite: Step $N = 10$			
Composite: Step $N = 20$			
Composite: Step $N = 50$			
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}:		
Adaptive 3/8 Simpsons Rule	N_{final}:		

For A = 10^{-5}

Exact Value (J_F): _____

Table 7: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$

Method	**N / Type**	$I_{\textbf{approx}}$ **(Approximate Value)**	$\lvert E \rvert$ **(Absolute Error)**
Simple Simpsons Rule			
Simple 3/8 Simpsons Rule			
Composite Integration (Fixed N)			
Composite: Step $N = 5$			
Composite: Step $N = 10$			
Composite: Step $N = 20$			
Composite: Step $N = 50$			
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}:		
Adaptive 3/8 Simpsons Rule	N_{final}:		

For A = 10^{-6}
Exact Value (J_F): _____

Table 8: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$

Method	**N / Type**	$I_{\textbf{approx}}$ **(Approximate Value)**	$\lvert E \rvert$ **(Absolute Error)**
Simple Simpsons Rule			
Simple 3/8 Simpsons Rule			
Composite Integration (Fixed N)			
Composite: Step $N = 5$			
Composite: Step $N = 10$			
Composite: Step $N = 20$			
Composite: Step $N = 50$			
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}:		
Adaptive 3/8 Simpsons Rule	N_{final}:		

For A = 10^{-7}
Exact Value (J_F): _____

Table 9: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$

Method	**N / Type**	$I_{\textbf{approx}}$ **(Approximate Value)**	$\lvert E \rvert$ **(Absolute Error)**
Simple Simpsons Rule			
Simple 3/8 Simpsons Rule			
Composite Integration (Fixed N)			
Composite: Step $N = 5$			
Composite: Step $N = 10$			
Composite: Step $N = 20$			
Composite: Step $N = 50$			
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}:		
Adaptive 3/8 Simpsons Rule	N_{final}:		

For A = 10^{-8}
Exact Value (J_F): _____

Table 10: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$

Method	**N / Type**	$I_{\textbf{approx}}$ **(Approximate Value)**	$\lvert E \rvert$ **(Absolute Error)**
Simple Simpsons Rule			
Simple 3/8 Simpsons Rule			
Composite Integration (Fixed N)			
Composite: Step $N = 5$			
Composite: Step $N = 10$			
Composite: Step $N = 20$			
Composite: Step $N = 50$			
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}:		
Adaptive 3/8 Simpsons Rule	N_{final}:		

For A = 10^{-9}
Exact Value (J_F): _____

Table 11: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$

Method	**N / Type**	$I_{\textbf{approx}}$ **(Approximate Value)**	$\lvert E \rvert$ **(Absolute Error)**
Simple Simpsons Rule			
Simple 3/8 Simpsons Rule			
Composite Integration (Fixed N)			
Composite: Step $N = 5$			
Composite: Step $N = 10$			
Composite: Step $N = 20$			
Composite: Step $N = 50$			
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}:		
Adaptive 3/8 Simpsons Rule	N_{final}:		

For A = 10^{-10}

Exact Value (J_F): _____

Table 12: Results for Locally Unsmooth Function $f(x) = \sqrt{x^2 + A}$

Method	**N / Type**	$I_{\mathbf{approx}}$ **(Approximate Value)**	$\lvert E \rvert$ **(Absolute Error)**
Simple Simpsons Rule			
Simple 3/8 Simpsons Rule			
Composite Integration (Fixed N)			
Composite: Step $N = 5$			
Composite: Step $N = 10$			
Composite: Step $N = 20$			
Composite: Step $N = 50$			
Adaptive Integration (Tolerance $\epsilon = \delta = 10^{-10}$)			
Adaptive Simpsons Rule	N_{final}: _____		
Adaptive 3/8 Simpsons Rule	N_{final}: _____		